

My Spring Project Features

Project Overview

It allows users to create auctions, place bids, track their activity, and determine winners automatically. It also gives administrators tools to manage users, oversee auctions, and review a complete audit trail.

The system supports both public browsing and protected operations through JWT authentication.

Main Capabilities

- User registration and login with JWT authentication.
 - Secure BCrypt password hashing.
 - User and administrator roles.
 - Public auction browsing, filtering, searching, sorting, and pagination.
 - English auctions with visible competitive bidding.
 - Sealed-bid auctions with private bids until closing.
 - Automatic auction opening and closing based on time.
 - Manual open, close, and cancellation operations.
 - Automatic winner determination.
 - Concurrency-safe bidding using a database row lock.
 - Personal views for created auctions, placed bids, and won auctions.
 - Administrator user management and platform-wide auction search.
 - Complete audit logging for important actions.
 - Consistent JSON error responses and request validation.
 - Swagger/OpenAPI documentation.
 - Oracle database support through Docker.
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Auction Formats

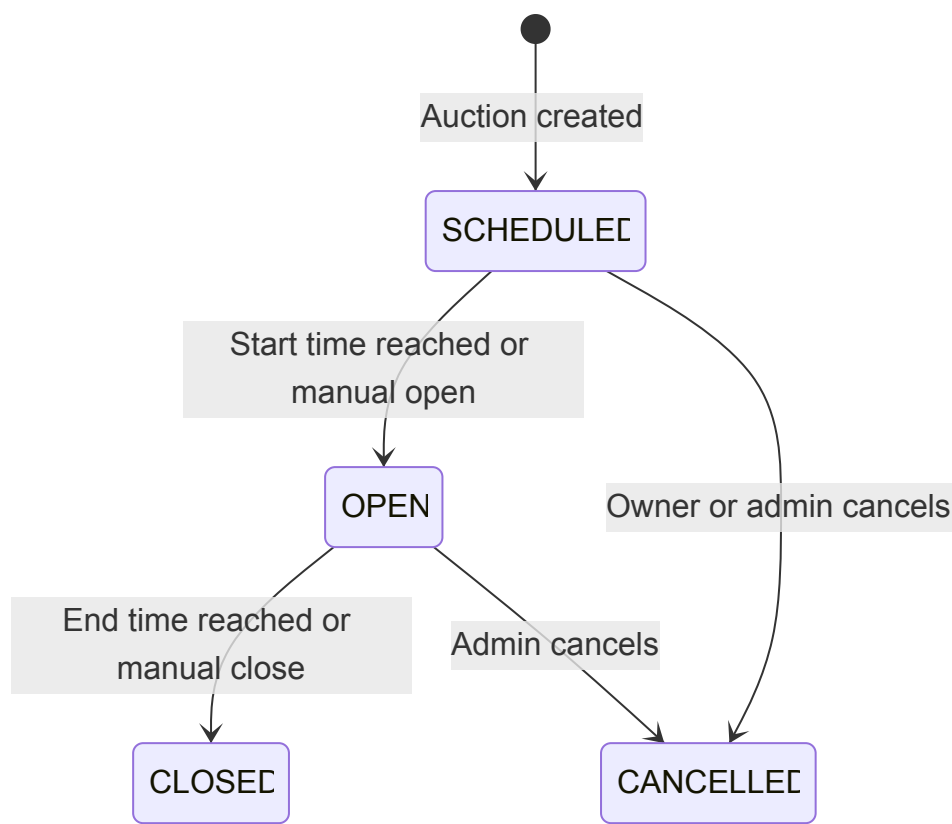
English Auction

- Everyone can see the bidding history and current highest bid.
- The first bid must be at least the starting price.
- Every later bid must beat the current highest bid by the minimum increment.
- A bidder may place multiple bids.
- The highest bid wins.

Sealed-Bid Auction

- Each user may submit only one final bid.
 - Bidders cannot see other users' bids before the auction closes.
 - A logged-in bidder may view only their own bid while the auction is still active.
 - After a successful close, all bids are revealed.
 - The highest bid wins.
 - If two bids have the same amount, the earliest bid wins.
 - A cancelled sealed auction remains private.
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Auction Lifecycle



Lifecycle Rules

- New auctions begin as `SCHEDULED`.
- Only scheduled auctions can be edited.
- Auctions can be opened manually by the owner or an administrator.
- The scheduler automatically opens auctions when their start time arrives.
- Open auctions can be closed manually by the owner or an administrator.
- The scheduler automatically closes auctions after their end time.
- A scheduled auction can be cancelled by its owner or an administrator.
- Only an administrator can cancel an already open auction.
- `CLOSED` and `CANCELLED` are terminal states.
- Clients cannot directly assign an auction status.

Automatic Winner Selection

When an auction closes, BidForge:

1. Finds the highest bid.
2. Uses the earliest bid as the winner when amounts are tied.
3. Creates an immutable auction-result record.
4. Records the winner, winning bid, final price, and closing time.
5. Creates a result even if the auction received no bids.

The close operation, result creation, and audit records are completed in one transaction.

Scheduler

A background scheduler runs approximately every ten seconds.

It:

- Finds `SCHEDULED` auctions whose start time has arrived and opens them.
- Finds `OPEN` auctions whose end time has passed and closes them.
- Uses the same service logic as manual open and close operations.
- Processes each auction independently so one failure does not stop the entire scheduler run.
- Records automatic actions under the actor name `SYSTEM`.

Concurrency Protection

BidForge protects state-changing auction operations with a pessimistic database row lock.

This is especially important when two bids arrive at nearly the same time:

- The first transaction locks the auction row.
- The second transaction waits.
- The first bid is validated and committed.
- The second bid then sees the updated highest bid and is revalidated correctly.

This prevents duplicate closing, stale bid validation, and corrupted auction state.

User Capabilities

An authenticated user can:

- View their own profile.
- Create an auction.
- Browse and search auctions.
- View auction details.
- View their own created auctions.
- View auctions they have won.
- Edit their own scheduled auction.
- Open or close their own auction when its state allows it.
- Cancel their own scheduled auction.
- Bid on another user's open auction.
- View their own bid history.
- View public English-auction bids.
- View only their own sealed bid before closing.

A seller cannot bid on their own auction.

Administrator Capabilities

An administrator can:

- Perform normal authenticated-user operations.
- List all user accounts.
- Enable or disable another user.
- Search auctions across the entire platform.
- Filter auctions by seller username.
- Open or close any auction when the state allows it.
- Cancel scheduled or open auctions.
- Read and filter the audit trail.

Important restrictions:

- An administrator cannot disable their own account.
 - Administrator status does not automatically allow editing another seller's auction.
 - Administrators must still follow normal auction-state and bidding rules.
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Public Capabilities

A visitor without an account can:

- Register.
- Log in.
- Browse, filter, search, sort, and paginate auctions.
- View auction details.
- View English-auction bid history.
- View all sealed bids after the auction closes.
- Access Swagger/OpenAPI documentation.

Protected operations require a valid JWT belonging to an enabled account.

API Endpoints

Authentication and Profile

Method	Endpoint	Access	Purpose
POST	/api/auth/register	Public	Register a normal user account.
POST	/api/auth/login	Public	Authenticate and receive a JWT.
GET	/api/users/me	Authenticated	View the logged-in user's profile.

Auctions

Method	Endpoint	Access	Purpose
POST	/api/auctions	Authenticated	Create a scheduled auction.
GET	/api/auctions	Public	Browse and search auctions.
GET	/api/auctions/mine	Authenticated	View auctions created by the logged-in user.
GET	/api/auctions/won	Authenticated	View auctions won by the logged-in user.
GET	/api/auctions/{id}	Public	View full auction details and result when closed.
PUT	/api/auctions/{id}	Owner	Edit a scheduled auction.
POST	/api/auctions/{id}/open	Owner or admin	Open a scheduled auction.
POST	/api/auctions/{id}/close	Owner or admin	Close an open auction and determine the winner.
POST	/api/auctions/{id}/cancel	Owner or admin	Cancel an auction according to state rules.

Auction Search Parameters

GET /api/auctions supports:

- status
- type
- category
- q for title search
- page
- size
- sort

Example:

```
GET /api/auctions?status=OPEN&type=ENGLISH&category=ART&q=painting&page=0&size=20&sort=endTime,asc
```

Bids

Method	Endpoint	Access	Purpose
POST	/api/auctions/{auctionId}/bids	Authenticated	Place a bid.
GET	/api/auctions/{auctionId}/bids	Public with visibility rules	View auction bids.
GET	/api/bids/my	Authenticated	View bids placed by the logged-in user.

Bid Visibility

Situation	Anonymous Visitor	Logged-In User
English auction	All bids	All bids
Active sealed auction	No bids	Only their own bid
Closed sealed auction	All bids	All bids
Cancelled sealed auction	No bids	Only their own bid

Administration

Method	Endpoint	Access	Purpose
GET	/api/admin/users	Admin	List all users.
PATCH	/api/admin/users/{id}/status	Admin	Enable or disable a user.
GET	/api/admin/auctions	Admin	Search all auctions, including by seller.
GET	/api/admin/audit-events	Admin	Read and filter the audit trail.

API Documentation

Method	Endpoint	Access	Purpose
GET	/swagger-ui.html	Public	Open Swagger UI.
GET	/swagger-ui/**	Public	Swagger UI resources.
GET	/v3/api-docs/**	Public	OpenAPI specification.

Validation and Error Handling

BidForge validates requests at multiple levels:

- DTO validation for required fields, formats, positive amounts, and future times.
- Service validation for auction-state and bidding rules.
- Database constraints for uniqueness, relationships, and required values.

Every error follows one consistent JSON structure containing:

- Timestamp
- HTTP status
- Error code
- Human-readable message
- Request path
- Optional field-validation errors

Audit Trail

Important activity is recorded in an insert-only audit log, including:

- User registration.
- User enable/disable changes.
- Auction creation and editing.
- Auction opening, closing, and cancellation.
- Bid placement.
- Winner selection.
- Scheduler actions performed by `SYSTEM`.

Administrators can filter the audit trail by actor, entity type, and entity ID.

Data and API Design

- Entities represent Oracle database tables.
 - DTOs define the public request and response formats.
 - Entities are never returned directly from the API.
 - Mappers convert entities into safe DTO responses.
 - Monetary values use `BigDecimal` for exact calculations.
 - Timestamps use UTC `Instant` values.
 - Lazy database relationships avoid unnecessary data loading.
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